SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0510 Slot A

COURSE OUTCOME COVERED
CO4: ***Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*** **Assessment Tool Weightage: 40%**

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Industry Problem Solving Task

Code of Conduct:

I D.HUSSAINAIAH(1924253000) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.HUSSAINAIAH

Rubrics for Evaluation

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	***Marks(100)***
Understanding of Industry Problem	20%	Comprehensive understanding of real world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation

Models available:

Viola-Jones: 78%, low latency

YOLO: 92%, moderate latency

Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

Answer:

This is a constraint-satisfaction problem rather than a plain 'pick the best accuracy' problem. A model has to clear all three conditions — the latency ceiling, the accuracy floor, and the hardware limit — before it can be considered a valid candidate, so each option needs to be checked against every condition individually.

Model	**Accuracy (need ≥85%)**	**Latency (need ≤100 ms)**	**Hardware fit (moderate compute)**
Viola-Jones	78% — fails	Low — passes	Fits easily — passes
YOLO	92% — passes	Moderate — passes	Matches moderate compute — passes
Deep Learning	95% — passes	High — fails	Needs more than moderate compute — fails

Viola-Jones is eliminated first: its 78% accuracy is below the 85% floor, so it never even gets to the latency check — an unacceptable accuracy rate rules it out on its own, regardless of how fast it runs.

Deep Learning looks attractive on paper because 95% is the highest accuracy of the three, but it fails two constraints at once: its high latency almost certainly pushes past the 100 ms ceiling, and its computational demand goes beyond what the 'moderate computation' hardware can realistically support. A model that can't run within the latency budget on the available hardware is not deployable here, no matter how accurate it is offline.

YOLO is the only model that clears every condition — accuracy above the 85% floor, latency within the 100 ms limit, and a computational footprint that matches moderate hardware. That combination is exactly what YOLO's single-stage detector architecture is designed for: it trades a small amount of peak accuracy for speed and efficiency, which is precisely the trade-off this problem is asking for.

Final decision: YOLO should be selected, since it is the only model that simultaneously satisfies the accuracy floor, the latency ceiling, and the hardware constraint.

2. Autonomous Vehicle Detection Logic

A vehicle detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms

Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

Answer:

Method	**Accuracy (need >90%)**	**Response time (need <50 ms)**
Method A	88% — fails	30 ms — passes
Method B	93% — passes	70 ms — fails
Method C	91% — passes	45 ms — passes

Method A is fast (30 ms), but its accuracy sits at 88% — below the 90% safety threshold. In an autonomous-vehicle context, accuracy directly maps to how reliably the system detects pedestrians, other vehicles, and obstacles, so falling short of the safety-critical accuracy bar disqualifies Method A even though its speed is excellent.

Method B clears the accuracy bar comfortably at 93%, but at 70 ms it responds too slowly — well past the 50 ms limit. At highway speeds, an extra 20 ms of detection delay translates into real distance the vehicle travels before reacting, which is not acceptable for a safety system.

Method C is the only option that satisfies both conditions together — 91% accuracy clears the safety threshold, and 45 ms keeps it inside the response-time budget. It doesn't have the single best score on either axis individually (Method B is more accurate, Method A is faster), but it's the only one that meets both requirements at once, which is what the problem actually demands.

Final decision: Method C should be selected, as it is the only method meeting both the accuracy requirement (>90%) and the response-time requirement (<50 ms) at the same time.

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

Answer:

The requirement isn't just 'accurate direction detection' in isolation — it's accurate direction detection under dense traffic specifically, and that second condition changes which method actually performs better in practice.

Optical Flow computes per-pixel motion vectors, which is exactly why it captures fine directional detail well in a clean scene. But dense traffic means many vehicles overlapping, occluding each other, and moving at different speeds close together — this is precisely the condition that makes Optical Flow noisy, because the assumptions it relies on (smooth, locally consistent motion) start breaking down when the scene is crowded. In other words, the accuracy advantage it has in ideal conditions degrades in exactly the scenario this system needs to handle.

Motion Estimation gives up some of that fine detail, but its stability is not scene-dependent in the same way — it continues to give a consistent, usable read on overall motion even when the scene gets busy. For a traffic monitoring system, a stable-but-slightly-coarser estimate of vehicle direction is far more useful than a highly detailed but noisy one, because the noise would translate into unreliable direction flags and false alarms exactly when the system is under the most load (peak traffic).

Final decision: Motion Estimation is more suitable for dense traffic, since its stability under crowding matters more here than Optical Flow's extra detail, which becomes unreliable in exactly this scenario.

4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain ≥80% accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

Answer:

Scenario	GMM Accuracy	Requirement (≥80%)	Result
Static scene	90%	≥80%	Meets requirement
Dynamic scene	65%	≥80%	Falls short by 15 points

The requirement is stated as ≥80% across all scenarios — meaning every scenario has to clear the bar, not just the average of them. GMM comfortably passes for static scenes at 90%, but in dynamic scenes it drops to 65%, well below the 80% floor. Since the standard is 'all scenarios', a single failing case is enough to say the system, as it stands, does not meet the requirement — even though it performs well half the time.

The 65% figure in dynamic scenes is consistent with a known limitation of plain GMM: constantly changing pixel regions (moving foliage, water, shifting light) get misclassified because the per-pixel Gaussian mixture can't distinguish 'expected background motion' from 'genuine foreground activity' without help.

So GMM alone is not sufficient here. The decision isn't to discard GMM entirely — its static-scene performance is solid and cheap to run — but to strengthen it specifically for the dynamic case: increase the number of Gaussian components with a tuned adaptive learning rate so recurring background motion is absorbed correctly, add morphological filtering to remove noise blobs, and/or pair it with a lightweight verification stage (e.g., a deep-learning detector) for dynamic regions only.

Final decision: GMM alone does not meet the requirement, since it fails the dynamic-scene case (65% < 80%). Recommendation — retain GMM for static regions and augment it with adaptive multi-Gaussian tuning plus a supplementary detection layer for dynamic regions.

5. Depth Estimation Strategy

Constraints:

- Only one camera available
- Moderate depth accuracy acceptable

Options:

Stereo Vision: high accuracy, needs two cameras

Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

Answer:

Approach	**Accuracy**	**Camera requirement**	**Feasible with 1 camera?**
Stereo Vision	High	Needs two cameras	Not feasible — hardware constraint violated
Structure from Motion	Moderate (acceptable per requirement)	Works with a single camera	Feasible — matches hardware and accuracy requirement

This question is really a hardware-availability gate before it's an accuracy question. Stereo Vision would give the best depth accuracy, but it fundamentally depends on two synchronized cameras to triangulate depth from the disparity between views — that's not an optimization detail, it's a hardware requirement the setup here can't meet with only one camera available. So Stereo Vision is ruled out immediately, regardless of how good its accuracy is, because the hardware constraint makes it physically infeasible in this scenario.

Structure from Motion (SfM) estimates depth using multiple frames captured over time from a single moving camera, tracking how features shift across frames to reconstruct 3D structure. That matches the single-camera constraint exactly. Its accuracy is only moderate compared to stereo, but the problem explicitly states that moderate accuracy is acceptable — so the one downside SfM has is already within tolerance.

Final decision: Structure from Motion should be selected, as it is the only approach that is actually deployable with a single camera, and its moderate accuracy already satisfies the stated requirement.